

Module 05



Sensory Processing & AI Perception

*How the Brain Makes Sense of the World
— And How AI Tries to Do the Same*

Where We Are

- Module 02 Foundations — Thousand Brains, Predictive Coding
- Module 03 Brain Anatomy — LIF Neurons, Neural Coding
- Module 04 Spiking Neural Networks — STDP, Neuromorphic Hardware
-
- **Module 05 Sensory Processing & AI Perception**
 - How sensory systems work — and how they inspired AI
 -
- Module 06 Memory, Learning, and AI Architectures
-

Learning Objectives

By the
end of
this
module,
you will
be able
to:

- 1 Describe how biological sensory systems convert physical energy into neural representations
 - 2 Explain the visual cortex hierarchy (V1 → IT) and its relationship to CNN architecture
 - 3 Compare all five sensory systems, identifying shared organizational principles
 - 4 Analyze how biological sensing inspired CNNs, spectrograms, and event-based vision
 - 5 Evaluate strengths and limitations of biological vs. artificial perception
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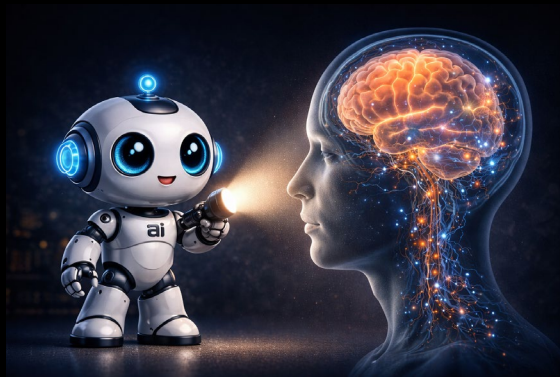
PART 1

How the Brain Perceives the World



Sensation, Perception, and the Five Senses

Your brain has never seen light.



It sits in complete darkness inside your skull.

All it receives are electrical signals — billions per second.

Yet from these signals alone, it constructs your entire world.

Everything you have ever seen, heard, touched, tasted, or smelled
was constructed by your brain from patterns of neural activity.

Sensation vs. Perception

	SENSATION	PERCEPTION
What	Raw transduction of physical energy → neural signals	Brain's constructed interpretation of those signals
Where	Sensory receptors (retina, cochlea, skin)	Cortical processing areas
Reality?	Relatively faithful to physical stimuli	Actively constructed — can be 'wrong' (illusions)
AI analogy	Camera pixel values / microphone waveform	Object recognition / speech understanding

Key insight: Perception is not passive reception. The brain actively constructs its best guess about the world from noisy, incomplete data + prior expectations.

Six Principles of Sensory Systems

All sensory systems share these organizational principles:

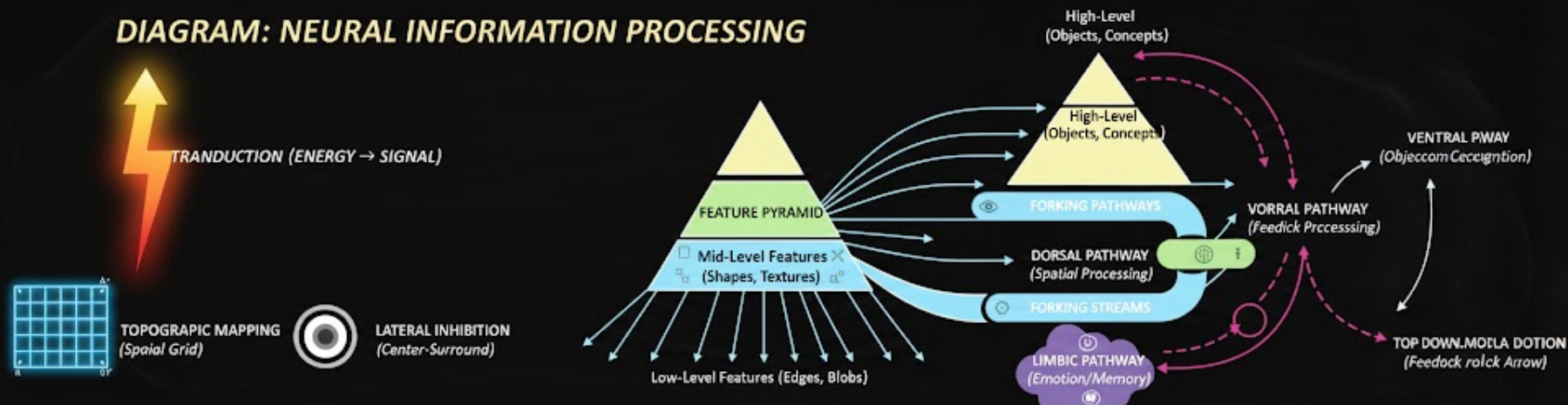
- 1 Transduction Physical energy $\rightarrow$ electrical signals
- 2 Hierarchical Processing Simple $\rightarrow$ complex $\rightarrow$ abstract features
- 3 Parallel Pathways 'What' and 'where' processed simultaneously
- 4 Topographic Mapping Spatial/frequency relationships preserved
- 5 Lateral Inhibition Neighbors suppress each other $\rightarrow$ sharpen contrasts
- 6 Top-Down Modulation Expectations shape what we perceive

AI status: Principles 1–5 well-implemented. Principle 6 (top-down) remains a frontier
attention mechanisms are the closest AI analog.



The Six Principles of Sensory Processing

DIAGRAM: NEURAL INFORMATION PROCESSING



The Retina: More Than a Camera

The retina is technically part of the brain — same embryonic tissue.

Photoreceptors:

Rods ~120M per eye — night vision, peripheral, B&W

Cones ~6M per eye — color, fine detail, fovea-concentrated

3 types: S (blue), M (green), L (red) → all color from combining these

The retina doesn't just sense — it computes.

Four additional cell types (horizontal, bipolar, amacrine, ganglion) form a local circuit that preprocesses visual information before it reaches the brain.

Most important computation: center-surround receptive fields →

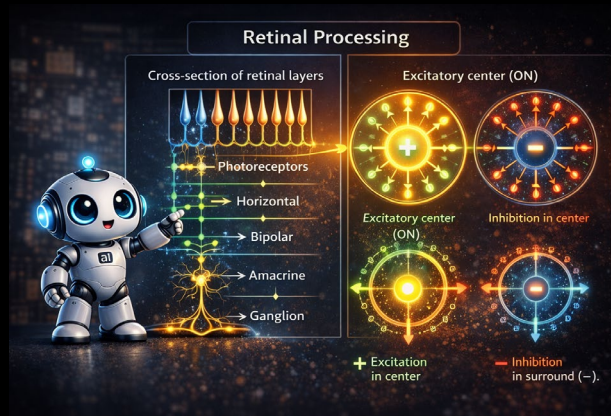
Center-Surround Receptive Fields

Each retinal ganglion cell responds to a small patch of the visual field:

- ON-center cells Excited by light in center, inhibited by light in surround
 - OFF-center cells Opposite pattern — excited by darkness in center
- **Result:** ganglion cells respond most strongly to edges and contrasts, not uniform illumination.

🗨️ → 🤖 AI Connection

- Center-surround $\approx$ Laplacian of Gaussian (LoG) edge detection
- CNN first layers independently learn center-surround-like filters
- DVS event cameras (Module 04) = silicon retinas: pixels fire on change



LGN: Gateway to the Cortex

Lateral Geniculate Nucleus — thalamic relay with 6 layers, 2 channels:

Magnocellular (M) pathway

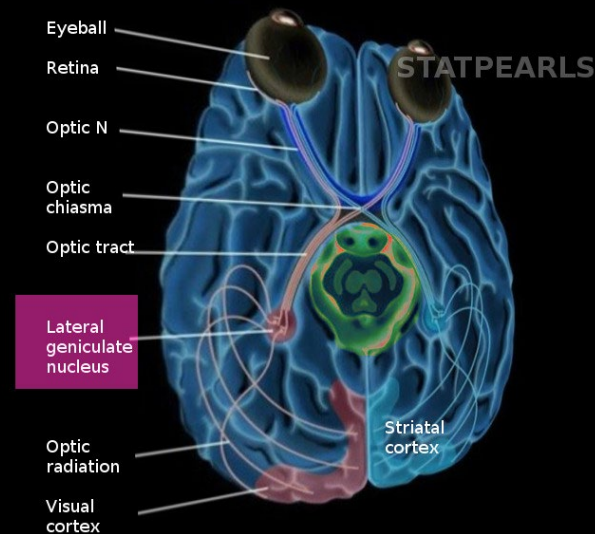
Large cells, fast • Motion, depth, low contrast
High temporal resolution, low spatial detail

Parvocellular (P) pathway

Small cells, slower • Color, fine detail, high contrast
High spatial resolution, slower temporal processing

Parallel pathways begin before visual cortex — Principle 3 in action

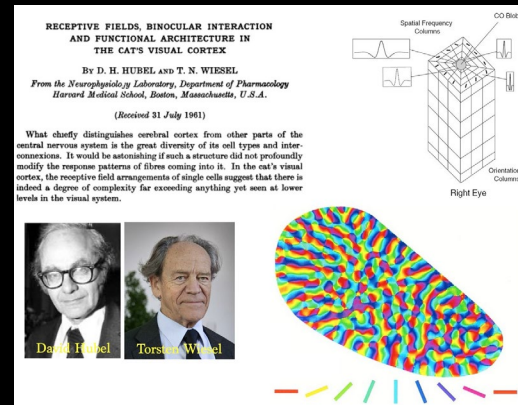
🧠 → 🤖 SlowFast networks (Feichtenhofer 2019) use one slow pathway for spatial detail and one fast pathway for motion — directly inspired by M/P channels.



Hubel & Wiesel: The Experiment That Changed AI

1958 — Johns Hopkins — Nobel Prize 1981

- Recorded from individual neurons in cat V1 while showing oriented bars.
 - Key discovery: V1 neurons are selective feature detectors.
 - Each neuron responds to a specific orientation of an edge at a specific location in the visual field.
 - Rotate the bar by 10–15° → neuron goes silent.
 - Move it to a different location → neuron goes silent.
 - This discovery — that the brain decomposes visual scenes into oriented edge features — directly led to the invention of CNNs.



Simple Cells, Complex Cells, Hypercomplex Cells

Simple cells

- Respond to oriented edges at a specific position
- Clear ON/OFF regions define preferred orientation

Complex cells

- Respond to oriented edges at any position within larger receptive field = Translation invariance!

Hypercomplex cells (end-stopped)

- Respond to edges of specific length — detect corners, endpoints

🧠 → 🤖 The CNN Mapping:

- Simple cells = Convolutional filters
- Complex cells = Pooling layers



The Visual Hierarchy: V1 → V2 → V4 → IT

V1 Oriented edges, spatial frequency

- Small receptive fields, precise location coding

V2 Contours, illusory contours, texture boundaries

- Begins integrating V1 features into meaningful groupings

V4 Shape, color, intermediate complexity

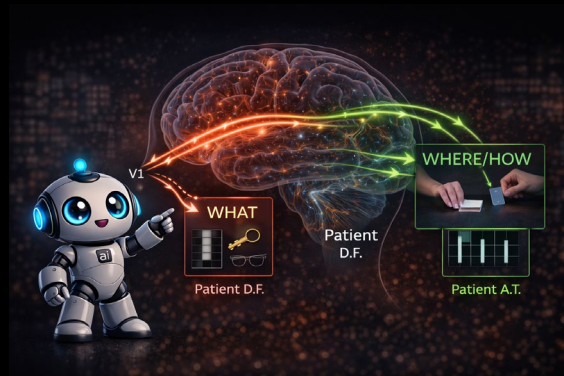
- Curves, angles, geometric shapes, color constancy

IT Whole objects, faces, scenes, identity

- Fusiform Face Area (FFA) for faces, PPA for places

The 'Jennifer Aniston neuron' (Quiroga 2005): individual neurons

- in IT respond to specific people — edges → identity in 4 processing stages.



Two Visual Streams: What vs. Where

Ventral stream — 'WHAT' pathway

V1 → V2 → V4 → IT → temporal cortex

Object identity: shape, face, color recognition

Damage → visual agnosia (can't recognize objects)

Dorsal stream — 'WHERE/HOW' pathway

V1 → MT/V5 → parietal cortex

Spatial processing: motion, depth, reaching, grasping

Damage → optic ataxia (can't reach for objects)

Double dissociation evidence:

Patient D.F. — can't recognize objects but grasps them correctly

Patient A.T. — recognizes objects but can't reach accurately

🧠 → 🤖 Most CV systems = ventral only. Robotics needs both streams.

Knowledge Check #1

- 1. What is the difference between sensation and perception? Give an example of each.
- 2. What type of receptive field do retinal ganglion cells have, and what computation does it perform?
- 3. Hubel & Wiesel found three cell types in V1. Name them and what each responds to.
- 4. Which V1 cell type = conv filters? Which = pooling layers?
- 5. A patient identifies objects but can't reach for them. Which stream is damaged?

The Auditory System: Sound to Signal

- Sound = pressure waves → eardrum → ossicles → cochlea
- The cochlea is the auditory retina.
- Inside: the basilar membrane vibrates at different positions for different frequencies:
 - Base (entrance) → high frequencies
 - Apex (tip) → low frequencies
 - = Tonotopic organization (Principle 4: topographic mapping)
- Hair cells sitting on the basilar membrane transduce vibration → neural signal.
- The cochlea performs a biological Fourier transform — decomposing
- complex sounds into frequency components, mechanically.



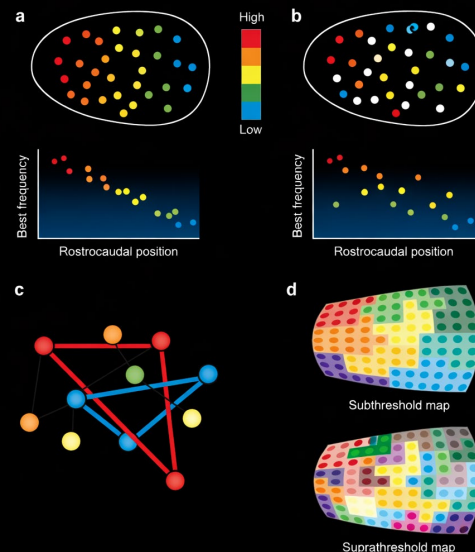
Tonotopy: The Auditory Equivalent of Retinotopy

Frequency mapping is preserved from cochlea all the way to cortex:

- Cochlea Position on basilar membrane = frequency
- Cochlear nucleus Tonotopic organization maintained
- Inferior colliculus Frequency maps sharpened
- A1 (primary auditory cortex) Frequency-tuned neurons in cortical columns

1. Retinotopy Nearby visual field → nearby neurons
2. Tonotopy Nearby frequencies → nearby neurons
3. Somatotopy Nearby body parts → nearby neurons

Same principle, different modalities — the brain reuses solutions.



The Auditory Pathway

Cochlea → Cochlear Nucleus → Superior Olivary Complex (SOC) → Inferior Colliculus → MGN (thalamus) → A1 (primary auditory cortex)

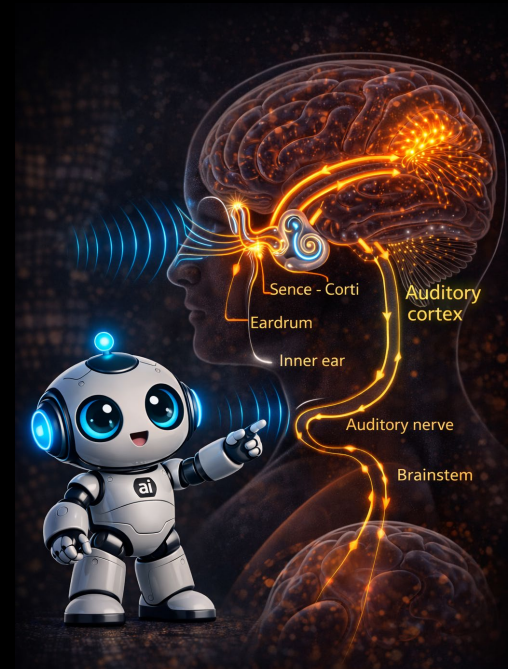
Key processing stations:

SOC First binaural comparison — compares timing and intensity between two ears for sound localization

A1 Frequency-tuned neurons organized in columns
Belt/parabelt areas process complex sounds, speech, music

Auditory 'what' vs. 'where' streams — parallel to vision!

🧠 → 🤖 Mel-frequency spectrograms in speech AI mirror tonotopic organization.



Visual vs. Auditory Systems

Feature	Vision	Audition
Stimulus	Electromagnetic radiation (light)	Pressure waves (sound)
Receptors	Rods & cones (retina)	Hair cells (cochlea)
Mapping	Retinotopy (spatial)	Tonotopy (frequency)
Thalamic relay	LGN	MGN
Primary cortex	V1 (occipital)	A1 (temporal)
Dual streams	Ventral (what) / Dorsal (where)	Anterior (what) / Posterior (where)
Strength	High spatial resolution	High temporal resolution

Same principles, different modalities — the brain reuses architectural patterns.

The Other Senses: Somatosensory System

Touch, temperature, pain, and proprioception (body position sense)

Four submodalities, four receptor types:

Mechanoreception Pressure, vibration, texture (Merkel, Meissner, Pacinian, Ruffini)

Thermoreception Warm and cool temperature channels

Nociception Pain — protective warning system

Proprioception Muscle spindles + joint receptors = body awareness

Somatotopic map → Sensory Homunculus

Distorted body map where fingertips and lips get huge cortical area

(more sensitive = more neurons allocated)



→ Haptic robotics: cobots use force/torque sensors for safe interaction.

Proprioception → joint encoders + IMUs in robotics.

Gustatory & Olfactory Systems

Gustatory (Taste)

5 basic tastes: sweet, salty, sour, bitter, umami

Each linked to survival: energy detection, mineral balance, spoilage warning,

toxin alert, protein finding

🧠 → 🤖 Electronic tongues for food quality control & pharma testing

Olfactory (Smell)

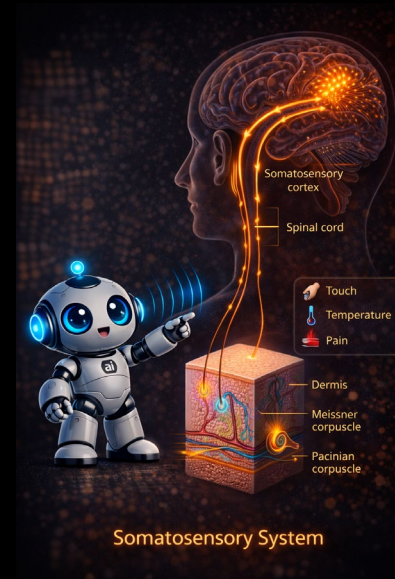
~400 receptor types → combinatorial coding → trillions of odors

Unique pathway: no thalamic relay — only sense with direct cortical access

Strong connection to emotion (amygdala) and memory (hippocampus)
→ explains why smells trigger vivid memories

🧠 → 🤖 Electronic noses detect disease biomarkers in breath.

Google (2023): ML model predicts odor from molecular structure.



Comparing All Five Senses

	Vision	Hearing	Touch	Taste	Smell
Energy	Light	Sound	Pressure	Chemical	Chemical
Receptors	Rods, cones	Hair cells	Mechano-receptors	Taste buds	Olfactory receptors
Mapping	Retinotopy	Tonotopy	Somatotopy	Taste map (debated)	Combinatorial coding
Thalamus?	LGN ✓	MGN ✓	VPL ✓	VPM ✓	No ✗
AI analog	CNNs, DVS	Spectrograms	Haptic robotics	E-tongues	E-noses

Olfaction is the only sense that bypasses the thalamus — direct cortical access explains the powerful smell-memory connection.

Same six principles apply to every modality — the brain reuses its best architectural ideas

Knowledge Check #2

1. What structure in the cochlea performs the auditory equivalent of a Fourier transform?
2. Name the three types of topographic mapping and which sensory system each belongs to.
3. Which is the only sense that bypasses the thalamus? Why might this matter for memory?
4. Both vision and hearing have dual 'what' and 'where' streams. What does this suggest about brain architecture?
5. Name one AI application inspired by each of the five senses.

PART 2

From Biology to AI Systems

Hubel & Wiesel → CNNs • Event-Based Vision • Bio-Inspired Pipelines

From Cat V1 to ImageNet: A Timeline

1962 Hubel & Wiesel

Simple/complex cells in cat V1 → oriented edge detectors

1980 Neocognitron (Fukushima)

First neural network with S-cells (conv) and C-cells (pooling)

Explicitly cited Hubel & Wiesel as inspiration

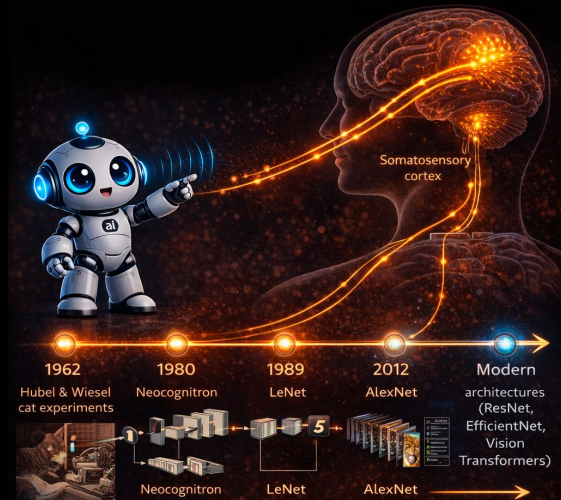
1989 LeNet (LeCun)

Added backpropagation training → first practical CNN

2012 AlexNet (Krizhevsky)

GPU-trained deep CNN wins ImageNet by 10+ points → deep learning revolution

50 years from cat neuroscience experiment to AI breakthrough.



Biology → CNN: Feature-by-Feature Mapping

Biology	CNN Equivalent	Match?
Simple cells	Convolutional filters	✓ Direct
Complex cells	Pooling layers	✓ Direct
V1 → V4 → IT hierarchy	Deep network layers	✓ Direct
Lateral inhibition	Batch norm / local response norm	~ Partial
Receptive field growth	Increasing RF with depth	✓ Direct
Recurrence / feedback	Not in standard CNNs	✗ Missing
Top-down modulation	Attention (partial)	~ Partial

What's different: biological vision uses massive recurrence (feedback loops) that standard CNNs lack. This is a key area of current research — recurrent CNNs and predictive coding networks.

CNN Feature Visualization: Seeing What Networks Learn

What CNNs Actually Learn (Zeiler & Fergus, 2014):

They visualized what each layer detects:

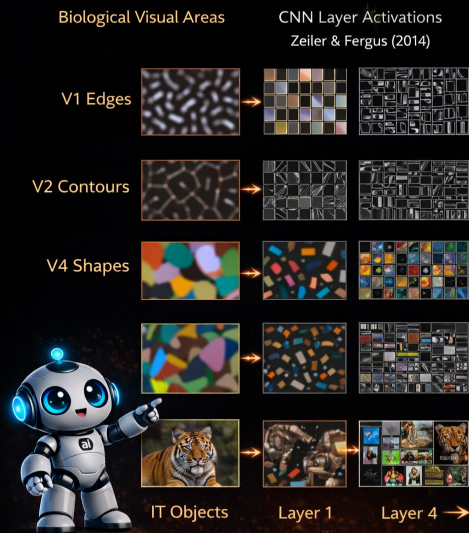
Layer 1: Edges, color blobs → matches V1

Layer 2–3: Textures, contours, corners → matches V2

Layer 4: Object parts, complex shapes → matches V4

Layer 5: Whole objects, faces, scenes → matches IT cortex

CNNs independently discovered the same hierarchy that evolution built into the visual cortex.



Event-Based Vision: Silicon Retinas

From Biological Retina to DVS (connects to Module 04)

Biological Retina: Ganglion cells respond to brightness *changes*, not absolute light ON cells fire for brighter, OFF cells fire for darker.

DVS Camera: Each pixel independently fires on brightness change Positive polarity = ON cell · Negative polarity = OFF cell.

The full bio-inspired pipeline:

DVS camera → SNN → Neuromorphic chip (silicon retina)
· (brain-like network) · (brain-like hardware)

Event-driven from sensor to decision. No frames, no clock.



Biological Retina → DVS: Feature Comparison

Feature	Biological Retina	DVS Camera
Response	To brightness changes	To brightness changes
Polarity	ON-center / OFF-center cells	Positive / negative events
Output	Spikes (asynchronous)	Events: (x, y, t, polarity)
Temporal res.	Milliseconds	Microseconds
Dynamic range	~100 dB	~120 dB
Power	~10 watts (whole brain)	~10-50 mW (sensor only)
Adaptation	Logarithmic light adaptation	Log intensity encoding

DVS cameras are the most literally bio-inspired sensor in all of AI — a silicon retina.

Event-Based Vision: Applications & Industry

Where DVS + SNN pipelines excel:

Autonomous vehicles High-speed obstacle detection, no motion blur

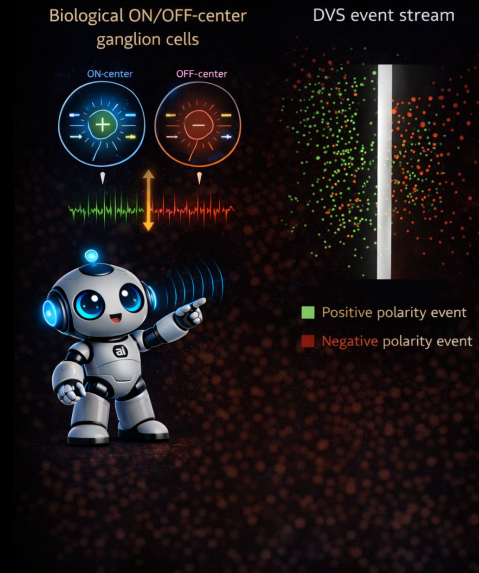
- Drone navigation Low-latency, low-power flight control
- Industrial inspection Defect detection at production speed
- Space applications Star tracking with extreme dynamic range

Industry players:

- Prophesee (Metavision sensors)
- iniVation (DAVIS cameras)
- Samsung (DVS research)
- Sony (event sensors in IMX products)

Career opportunity:

- DVS + SNN + neuromorphic chip expertise
- is a rare and valuable skill combination.



Knowledge Check #3

1. Name the four milestones from cat V1 experiments to the deep learning revolution.
2. Which CNN component corresponds to simple cells? Which to complex cells?
3. CNN feature visualization shows layers learn edges → textures → parts → objects. Which brain areas does this match?
4. How do DVS camera events correspond to biological retinal ganglion cell responses?
5. Name two features where biological vision still exceeds standard CNNs.

PART 3

Multimodal Integration & Frontiers

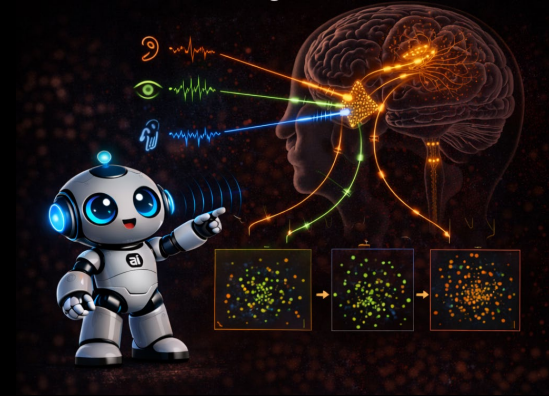
Combining the Senses • Biology vs. AI • What Comes Next

Multisensory Integration: Three Rules

Superior Colliculus: 3 Rules of Multisensory Integration

These neurons combine input from multiple senses following three rules:

- 1 • **Spatial Rule** Same location = boosted response See AND hear a bird in the same direction → stronger signal
- 2 • **Temporal Rule** Same time = integrated Lip movements must sync with speech (~200 ms window)
- 3 • **Inverse Effectiveness** Weak signals benefit most Quiet sound + dim flash → big boost Loud sound alone → adding another sense barely helps



The McGurk Effect: When Vision Changes Hearing

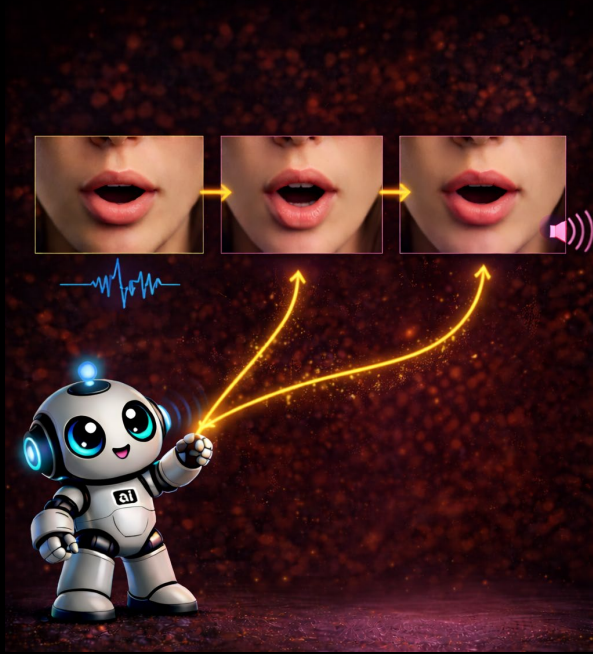
McGurk & MacDonald (1976) — one of the most striking perceptual illusions:

Audio: "ba" + Video: lips saying "ga" → You hear: "da"

Your brain doesn't pick one, it invents a third sound that was never there.

Why it works: Spatial Rule both signals come from the same face. Temporal Rule audio and lip movement are in sync. Result a fused percept neither sense produced alone

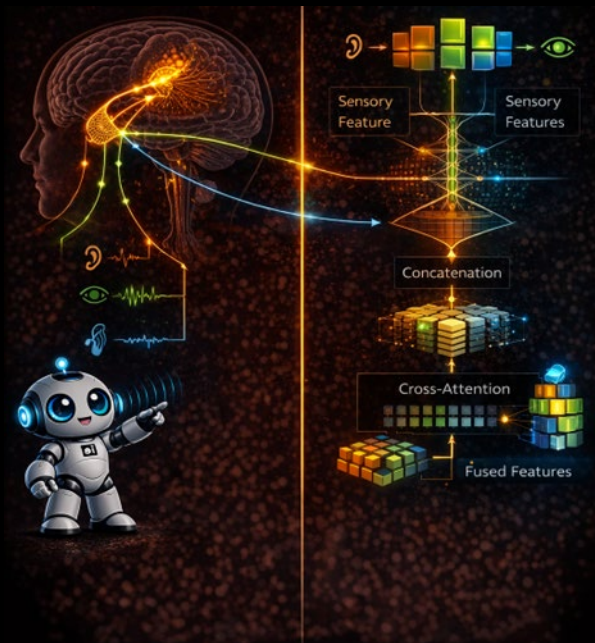
Try it yourself: Search "McGurk Effect" on YouTube. Close and open your eyes to hear it switch!



Multimodal AI: CLIP, GPT-4V, Sensor Fusion

How AI Handles Multimodal Integration

- **Vision + Language** CLIP aligns image and text embeddingsGPT-4V, Gemini process images + text in unified context
- **Audio + Visual Speech recognition + lip reading** → robust in noisy environments Multimodal emotion recognition (face + voice + text)
- **Sensor Fusion** (Autonomous Vehicles)Camera + LiDAR + radar + IMU → unified scene understanding
- **Biology vs AIBiology** uses principled rules (spatial, temporal, inverse effectiveness)AI mostly learns fusion from dataOpen question: could bio-inspired rules improve AI fusion? Opus 4.6Extended

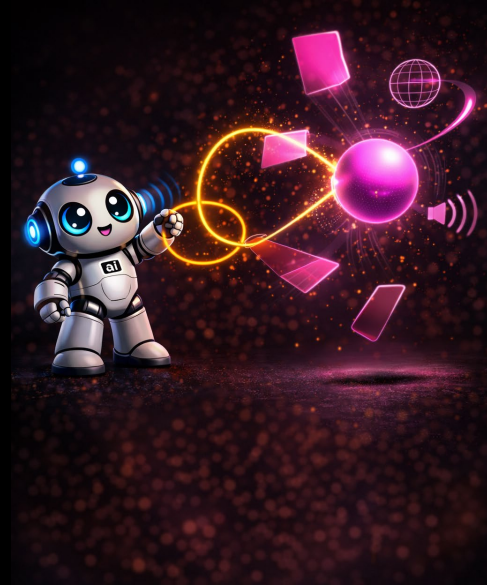


The Binding Problem

You see a red ball bouncing with a sound.
"Red" is in V4. "Shape" in IT. "Motion" in MT. "Sound" in auditory cortex.
Yet you experience one unified event. How?

Proposed Biological Solutions Temporal Synchrony: neurons for the same object fire in sync (~40 Hz gamma) Attention: selective focus binds features of the attended object Global Workspace: bound representations broadcast to consciousness

The AI Gap Current multimodal models concatenate or cross-attend They don't truly "bind" features into unified objects the way brains do This remains an unsolved problem in both neuroscience and AI



What Biology Does Better — And Where AI Excels

Biology Wins

- Robustness to noise & occlusion
- Energy efficiency (~10 watts)
- One-shot learning of new objects
- Context & common sense
- Invariance with minimal training
- Principled multimodal integration
- Top-down modulation

AI Wins

- Speed on massive datasets
- Consistency & repeatability
- Beyond-human spectrum sensing
- Scalability (1000s of sensors)
- No attentional bottleneck
- Precise numerical computation
- Rapid retraining / transfer

The convergence: bio-inspired AI (attention, event-driven processing, capsule nets) is closing the gap.

Knowledge Check #4

1. Name the three rules of multisensory integration and explain inverse effectiveness.
2. In the McGurk Effect, audio 'ba' + video 'ga' produces what percept? Why?
3. What is the binding problem? Has AI or neuroscience solved it?
4. Name two things biological perception does better than AI, and two where AI excels.

Lab Preview: Bio-Inspired Perception Pipeline



A Retinal Processing

Build center-surround DoG filters mimicking ganglion cells

B V1 Processing

Gabor filter bank at multiple orientations = simple cells

C CNN Comparison

Extract first-layer filters from VGG/ResNet, compare to Gabor bank

D Hierarchical Activation Maps

Trace early→mid→late layers, map to V1→V2→V4→IT

E Reflection

Where do bio filters match CNN filters? Where do they diverge?

Key Takeaways

1 Sensory systems share six organizational principles

Hierarchy, parallel pathways, topographic maps, lateral inhibition...

2 Visual cortex → CNNs is the clearest brain→AI success story

Hubel & Wiesel → Neocognitron → LeNet → AlexNet → modern DL

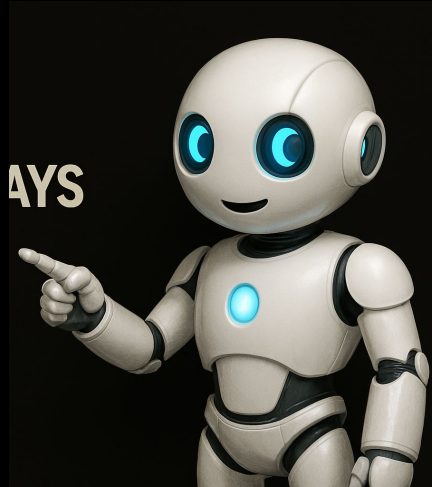
3 All five senses inspire AI applications

Vision→CNNs, hearing→spectrograms, touch→haptic robots,
taste→e-tongues, smell→e-noses

4 Multimodal integration remains a frontier

Biology uses principled rules; AI learns from data.

The binding problem is unsolved in both fields.



Bridge to Module 06

Module 05 gave you:

SENSING — how the brain converts physical energy to neural signals

HIERARCHY — V1 → IT progression and its CNN parallel

INTEGRATION — multimodal combining of senses

Module 06 will explore:

Memory, Learning, and AI Architectures

From perceiving the world → to remembering it.